



THE UNIVERSITY OF
MELBOURNE

ACCT90002

Financial Statement

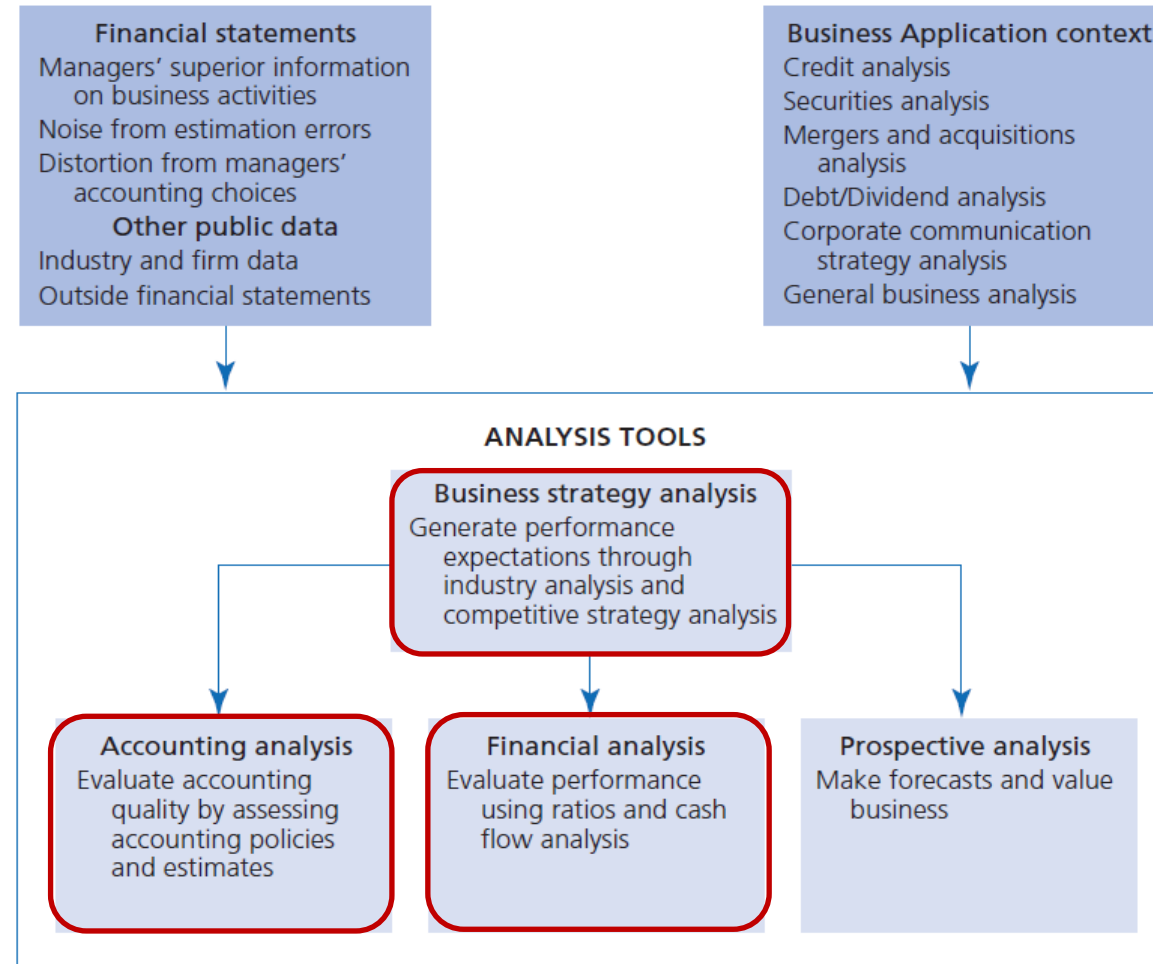
Analysis

Week 5: Financial analysis (1)



Framework - Financial Statement Analysis

FIGURE 1.3 Analysis using financial statements



Financial analysis

➤ There are two primary tools in financial analysis:

➤ **Ratio analysis** – to assess how various line items in financial statements relate to each other and to measure **relative performance**

➤ **Cash flow analysis** – to evaluate **liquidity** and the management of **operating**, **investing**, and **financing** activities as they relate to cash flow

Determinants of Firm Value and Ratio Analysis

Profitability and growth drive firm value.

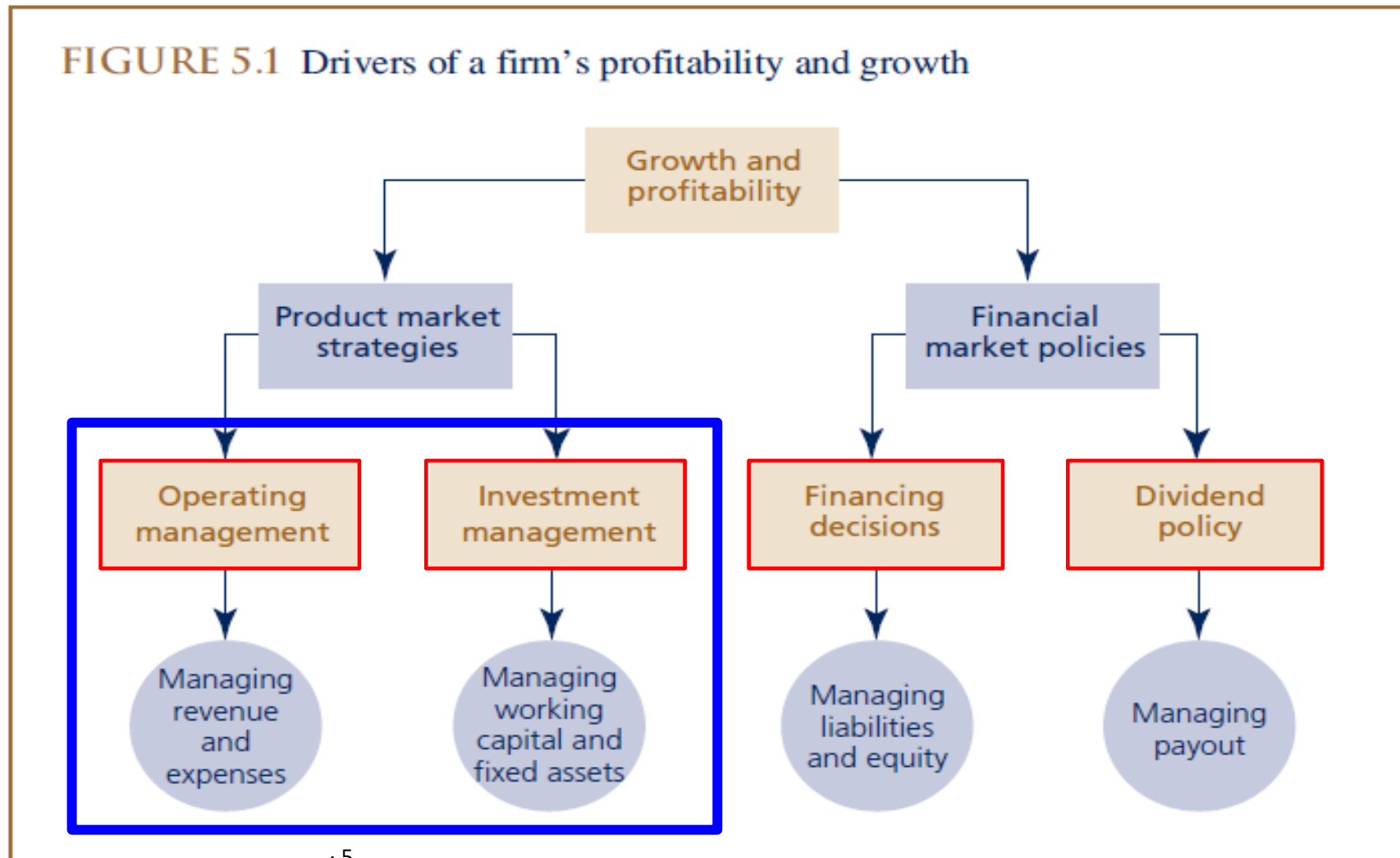
Managers can employ four levers to achieve growth and profit targets:

- Operating management
- Investment management
- Financing strategy
- Dividend policy

Ratio analysis seeks to evaluate the firm's effectiveness in these areas.

Drivers of profitability and growth

FIGURE 5.1 Drivers of a firm's profitability and growth



Ratio Analysis

Evaluating ratios requires comparison against some benchmarks, such as:

- Ratios over time from prior periods (**time series**)
- Ratios of other firms in the industry (**cross-sectional**)
- Some absolute benchmark
- Common size analysis

Effective ratio analysis must attempt to relate underlying business factors to the financial numbers.

Measuring overall profitability

➤ Return on Equity (ROE) is a comprehensive measure of and is a good starting point to ***systematically*** analyze firm performance

$$\text{ROE} = \frac{\text{Net income} - \text{preferred dividends}}{\text{Average Shareholders' equity}}$$

TABLE 5.1 Return on equity for Bega Cheese Group (Bega) and Warrnambool Cheese and Butter Factory (WCBF)

Data items and ratio	Bega 2013	Bega 2012	WCBF 2013	WCBF 2012
Profit for the year or net income (\$'000)	25 445	20 429	7 490	15 184
Average shareholders' equity (\$'000)	254 196	213 149.5	160 999.5	155 971.5
Return on equity	10.01%	9.58%	4.65%	9.74%

Time-series analysis

Measuring overall profitability

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Return on equity	10.01%	9.58%	4.65%	9.74%

Cross-sectional analysis

Return on Equity (ROE)

- How well managers are employing **the funds invested by the firm's shareholders** to generate returns
- **Average long-run** ROE for Australian listed firms: 10-12%
- In the **long run**, the value of a firm's equity is determined by the relationship between its **ROE** and its **cost of equity**

Decomposing ROE : Traditional approach



Decomposing ROE : Traditional approach

➤ watch a video on DuPont Analysis (<http://go.unimelb.edu.au/h9ij>)

➤ ROE is affected by **two factors**:

- How profitably the firm employs its assets;
- How big the firm's asset base is relative to shareholders' investment

$$\frac{\text{Net profit}}{\text{Equity}} = \frac{\text{Net profit}}{\text{Assets}} * \frac{\text{Assets}}{\text{Equity}}$$

$$\text{ROE} = \text{ROA} * \text{Financial leverage (Equity Multiplier)}$$

ROA tells us how much profit a company is able to generate for each dollar of assets invested.

Financial leverage (the Equity Multiplier) indicates how many dollars of assets the firm is able to deploy for every dollar invested by shareholders.

Decomposing ROE : Traditional approach (Cont'd)

- **ROA = Net profit margin (return on sales) * Asset turnover**

$$\frac{\text{Net profit}}{\text{Assets}} = \frac{\text{Net profit}}{\text{Sales}} * \frac{\text{Sales}}{\text{Assets}}$$

- **Net profit margin** tells us how much the company is able to keep as profits for each dollar of sales it makes

- **Asset turnover** indicates how many dollars of sales the firm is able to generate for each dollar of assets

- **Total decomposition of ROE (DuPont Equation):**

$$\text{ROE} = \frac{\text{Net profit}}{\text{Sales}} * \frac{\text{Sales}}{\text{Assets}} * \frac{\text{Assets}}{\text{Equity}}$$

Decomposition helps in answering questions about the source of a firm's profitability

FIGURE 5.4 Traditional decomposition of ROE

Ratio	Kathmandu		Super Retail Group		Other peers	
	2018	2017	2018	2017	2018	2017
Net profit margin (ROS)	10.2%	8.5%	4.9%	4.1%	5.8%	6.0%
× Asset turnover	0.94	1.00	1.55	1.58	1.54	1.87
= Return on assets	9.6%	8.6%	7.7%	6.4%	8.8%	11.2%
× Leverage (assets / equity)	1.41	1.39	2.14	2.10	1.87	2.53
= Return on equity	13.5%	11.9%	16.4%	13.5%	16.5%	28.4%

Drawbacks with the traditional approach:

$$\text{ROE} = \frac{\text{Net profit}}{\text{Sales}} * \frac{\text{Sales}}{\text{Assets}} * \left(1 + \frac{\text{Liabilities}}{\text{Equity}} \right)$$

- **ROA denominator** includes the assets claimed by *all* providers of capital to the firm, whereas **ROA numerator** includes *only* the earnings available to *equity holders* (*a mismatch of measurement scope*)
- **Assets** include both **operating** as well as **investment** assets (*no differentiation by return rates*)
- **Net profit** includes **profit from operating activities**, as well as **interest income** and **interest expense**, which are consequences of **investment** and **financing activities** (*no differentiation by sources*)

Drawbacks with the traditional approach:

$$ROE = \frac{\text{Net profit}}{\text{Sales}} * \frac{\text{Sales}}{\text{Assets}} * \left(1 + \frac{\text{Liabilities}}{\text{Equity}} \right)$$

- Fail to differentiate **interest-bearing liabilities** (e.g., debt) from **non-interest-bearing (operating) liabilities** (e.g., provisions)
- Fail to recognize '**negative debt**' (i.e., cash and marketable securities)
- **Financial leverage ratio:** A double edged sword! (*fail to differentiate direct positive effect on ROE from indirect negative effect through increasing financial risk and borrowing costs*)

Decomposing ROE : Alternative approach



Reformulating the balance sheet and income statement

Assets = Liabilities + SH Equity

CA + LA = CL + LL + SH Equity

- **Minus cash and marketable securities**

CA + LA – C = CL + LL – C + SH Equity

- **Minus non-interest bearing liabilities (including short-term and long-term)**

CA + LA – C – non-interest bearing liabilities = CL + LL – C – non-interest bearing liabilities + SH Equity

OWC = CA – C – non-interest bearing ST liabilities

NLTA = LA – non-interest bearing LT liabilities

ND = CL + LL – non-interest bearing liabilities – C

OWC + NLTA = ND + SH Equity



Reformulating the balance sheet and income statement

➤ Net interest expense after tax (NIEAT):

- $(\text{Interest Expense} - \text{Interest Income}) * (1 - \text{Tax Rate})$

➤ Net operating profit after taxes (NOPAT):

- Net income + Net interest expense after tax

➤ Operating working capital (OWC):

- $(\text{CA} - \text{Cash and marketable securities}) - (\text{CL} - \text{Short term debt})$

➤ Net long-term (non-current) assets (NLTA):

- Total long term assets – non-interest bearing long term liabilities (e.g., tax liabilities, P-E benefit obligations, provisions, etc.)

➤ Net debt:

- Total interest bearing liabilities – cash and marketable securities

➤ Net (operating) assets: $\text{OWC} + \text{NLTA}$

➤ Net capital: Net debt + shareholders' equity

$$\begin{aligned} \text{➤ Net (operating) assets} &= \text{Net debt} + \text{Shareholders' Equity} \\ \text{OWC} + \text{NLTA} &= \text{Net debt} + \text{Shareholders' Equity} \end{aligned}$$



Condensed Balance Sheet

**Net (operating) assets= Net debt +
Shareholders' Equity**

**OWC + NLTA = Net debt +
Shareholders' Equity**

	\$
Operating working capital (OWC)	1000
Net long-term (non-current) assets (NLTA)	<u>5500</u>
Net (operating) assets	<u>6500</u>
 Net debt	 2500
Shareholders' equity	<u>4000</u>
 Net capital	 <u>6500</u>

Condensed Income Statement

	\$
Sales	5000
NOPAT	450
Net interest expense after tax	(150)
Net income	300

Consolidated statement of financial position

at 31 March

	Note	2012 £m	2011 £m
Non-current assets			
Goodwill	9	38,350	45,236
Other intangible assets	9	21,164	23,322
Property, plant and equipment	11	18,655	20,181
Investments in associates	14	35,108	38,105
Other investments	15	791	1,381
Deferred tax assets	6	1,970	2,018
Post employment benefits	23	31	97
Trade and other receivables	17	3,482	3,877
		119,551	134,217
Current assets			
Inventory	16	486	537
Taxation recoverable		334	281
Trade and other receivables	17	10,744	9,259
Other investments	15	1,323	674
Cash and cash equivalents	18	7,138	6,252
		20,025	17,003
Total assets		139,576	151,220
Equity			
Called up share capital	19	3,866	4,082
Additional paid-in capital		154,123	153,760
Treasury shares		(7,841)	(8,171)
Retained losses		(84,184)	(77,661)
Accumulated other comprehensive income		10,971	15,545
Total equity shareholders' funds		76,935	87,555
Non-controlling interests		2,090	2,880
Put options over non-controlling interests		(823)	(2,874)
Total non-controlling interests		1,267	6
Total equity		78,202	87,561
Non-current liabilities			
Long-term borrowings	22	28,362	28,375
Taxation liabilities		250	350
Deferred tax liabilities	6	6,597	6,486
Post employment benefits	23	337	87
Provisions	24	479	482
Trade and other payables	25	1,324	804
		37,349	36,584
Current liabilities			
Short-term borrowings	22	6,258	9,906
Taxation liabilities		1,898	1,912
Provisions	24	633	559
Trade and other payables	25	15,236	14,698
		24,025	27,075
Total equity and liabilities		139,576	151,220

31 March, 2012

Operating working capital (OWC):

Current assets **20025**

Less:

Cash and marketable securities **8461**

(cash and cash equivalent:

7138 + other investments: 1323)

11564

Current liabilities **24025**

Less:

Interest-bearing current liabilities **6258**

(short-term borrowings: 6258)

17767

OWC **-6203**

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Total equity and liabilities		139,576	151,220

31 March, 2012

Net long-term (non-current) assets:

Total non-current assets **119551**

Less:

Non-interest bearing non-current liabilities **8987**

(total non-current liabilities: 37349

– long-term borrowings: 28362)

Net long-term (non-current) assets **110564**

Net (operating) assets **104361**

Consolidated statement of financial position

at 31 March

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Trade and other payables	25	15,236	14,698
		24,025	27,075
Total equity and liabilities		139,576	151,220

31 March, 2012

Shareholders equity (given)

78202

Net debt:

Long-term debt

28362

Short-term borrowings

6258

Less:

Cash and marketable securities

8461

Net debt

26159

Net debt + SH equity

104361

Decomposing ROE – an alternative approach

ROE (Return on Equity) = Net income/Equity

= NOPAT/Equity – (Net interest expense after tax)/Equity

= NOPAT/Net operating assets * Net operating assets/Equity – Net interest expense after tax/Net debt * Net debt/Equity

= NOPAT/Net operating assets * (1 + Net debt/Equity) – Net interest expense after tax/Net debt * Net debt/Equity

Transformation highlighted in blue

Net operating assets = Net debt + Shareholders' Equity

1st decomposition level

$$\begin{aligned}
 & \text{Operating ROA} && \text{Net financial leverage} \\
 & \text{NOPAT/Net operating assets} * (1 + \text{Net debt/Equity}) \\
 & - \text{Net interest expense after tax/Net debt} * \text{Net debt/Equity} \\
 & \text{Effective interest rate after tax} && \text{Net financial leverage} \\
 & = \text{Operating ROA} + (\text{Operating ROA} - \text{Effective interest rate after tax}) \\
 & * \text{Net financial leverage} \\
 & \text{Spread}
 \end{aligned}$$

Financial Leverage Effect (gain/loss)

$$\text{ROE} = \text{Operating ROA} + \text{Spread} * \text{Net financial leverage}$$

- Operating ROA = NOPAT/Net operating assets
- Spread = Operating ROA – Effective interest rate after tax
- Net financial leverage = Net debt/Equity

2nd decomposition level

$$\text{Operating ROA} = \text{NOPAT} / \text{Net operating assets}$$

$$= \underbrace{\text{NOPAT} / \text{Sales}}_{\text{(i.e., NOPAT margin)}} * \text{Sales} / \text{Net operating assets}$$

Average operating ROA for large Australian firms over long term is 8 to 10%

FIGURE 5.6 Distinguishing operating and financing components in ROE decomposition

Ratio	Kathmandu		Super Retail Group	
	2018	2017	2018	2017
Net operating profit margin	10.3%	8.9%	5.4%	4.6%
× Net operating asset turnover	1.23	1.31	2.19	2.17
= Return on net operating assets	12.7%	11.6%	11.9%	9.9%
After tax cost of net debt (net interest expense after tax / net debt)	2.4%	6.4%	3.2%	3.1%
Spread (RNOA – after-tax cost of net debt)	10.3%	5.1%	8.7%	6.8%
× Financial leverage (net debt/equity)	0.08	0.07	0.52	0.53
= Financial leverage gain	0.8%	0.4%	4.5%	3.6%
ROE = return on operating assets + financial leverage gain	13.5%	11.9%	16.4%	13.5%

Evaluating financial management: Leverage

ROE = Operating ROA + Spread * Net financial leverage

- Operating ROA = NOPAT/Net assets
- Spread = Operating ROA – Effective interest rate after tax
- Net financial leverage = Net debt/Equity

➤ **Spread** = Operating ROA – Effective interest rate after tax

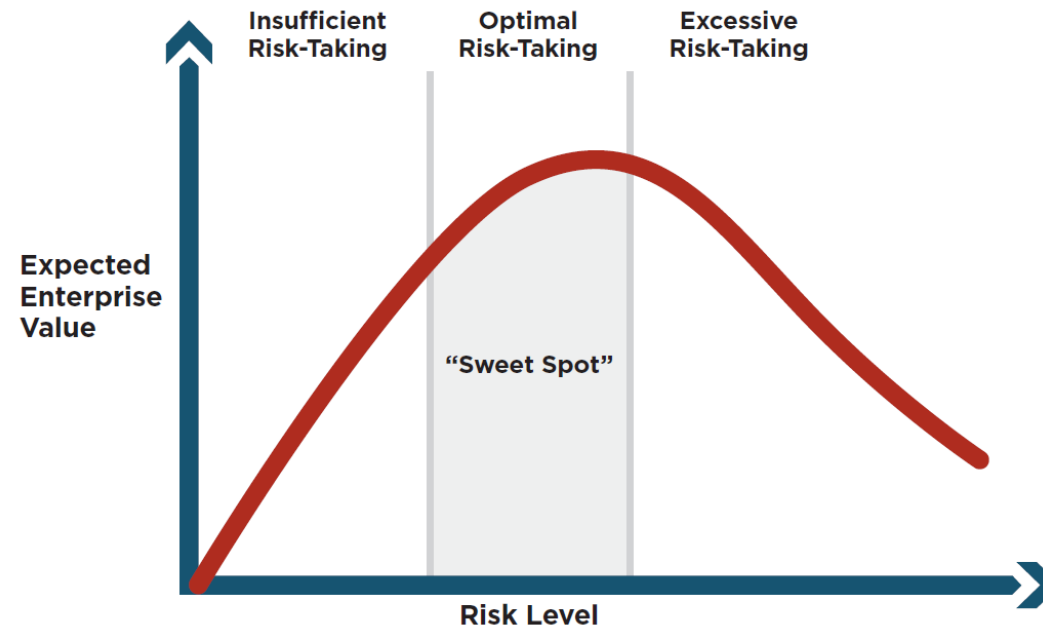
- Increases a firm's ROE as long as the cost of debt is less than the return from investing these funds
- Financial leverage can therefore benefit shareholders but also increases the risk of financial distress

Financial Leverage Effect

Financial leverage effect = **Spread** * **Net financial leverage**

➤ **Spread** = Operating ROA – Effective interest rate after tax

Increases a firm's ROE as long as the cost of debt is less than the return from investing these funds (Operating ROA)



Framework for alternative approach

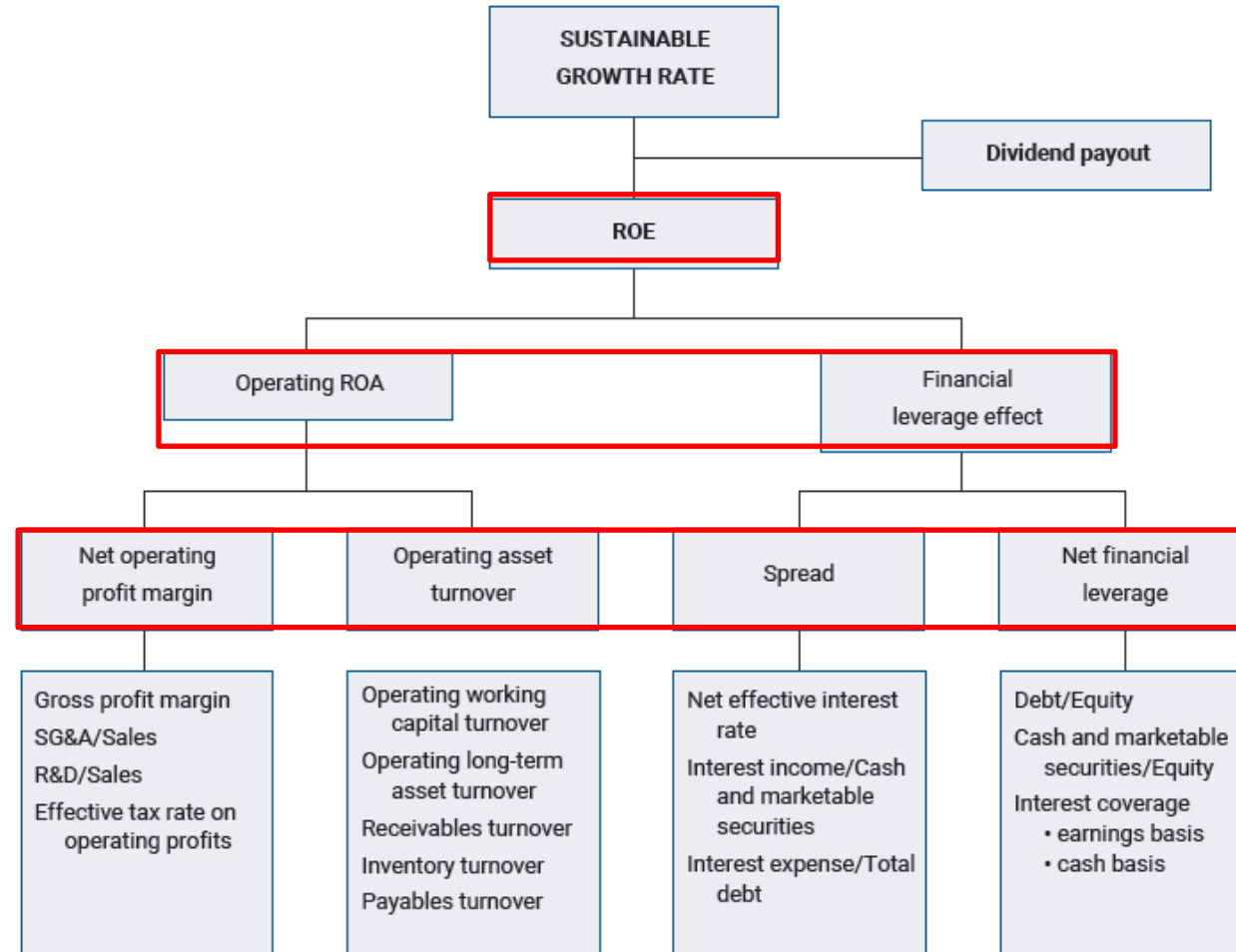


FIGURE 5.12 Sustainable growth rate framework for financial ratio analysis



Assessing Operating Management: income statement ratios

Common-sized income statements facilitate comparisons of key line items across time and different firms.

As well, the following ratios are helpful:

- Gross profit margin
- EBIT margin (Earnings before interest and tax)
- EBITDA margin (Earnings before interest, tax, depreciation and amortisation)

Gross profit margin

Measures the profitability of sales, less direct costs of sales:

$$\text{Gross profit margin} = \frac{\text{Revenue} - \text{Cost of sales}}{\text{Revenue}}$$

The gross profit margin is an indicator of:

- The price premium that a firm's product commands in the market
- The efficiency of a firm's procurement and/or production process

EBIT AND EBITDA margins

The EBIT margin provides a comprehensive measure of operations:

$$\text{EBIT margin} = \frac{\text{EBIT}}{\text{Revenue}}$$

The EBITDA margin eliminates the significant non-cash expenses of depreciation and amortisation along with interest and taxes:

$$\text{EBITDA margin} = \frac{\text{EBITDA}}{\text{Revenue}}$$

KATHMANDU'S COMMON-SIZED INCOME STATEMENT AND PROFITABILITY RATIOS

FIGURE 5.7 Common-sized income statement and profitability ratios

Ratio	Kathmandu		Super Retail Group		Other peers
	2018	2017	2018	2017	2018
Line item as a percentage of sales					
Sales	100%	100%	100%	100%	100%
Cost of sales	-36.6%	-38.0%	-55.1%	-55.3%	55.3%
Gross profit	63.4%	62.0%	44.9%	44.7%	44.7%
Other income	0.0%	0.0%	0.3%	0.1%	-2.7%
Selling expenses	-31.3%	-32.3%	-16.2%	-16.5%	
Administrative and general expenses	-14.1%	-13.8%	-21.5%	-21.8%	
Earnings before interest, tax, depreciation & amortisation	18.0%	15.9%	11.5%	11.2%	4.9%
Depreciation and amortisation	-3.0%	-3.1%	-3.0%	-2.9%	2.1%
EBIT	15.0%	12.8%	8.5%	8.4%	2.8%
Finance income	0.0%	0.0%	0.0%	0.0%	0.1%
Finance expenses	-0.2%	-0.5%	-0.7%	-0.7%	1.5%
Finance costs – net	-0.2%	-0.5%	-0.7%	-0.7%	1.4%
Profit before income tax	14.8%	12.3%	6.9%	5.7%	7.6%
Income tax expense	-4.6%	-3.8%	-1.9%	-1.6%	2.4%
Profit after income tax	10.2%	8.5%	5.0%	4.1%	2.6%

Evaluating Investment Management: Decomposing Asset Turnover

Asset management is a key indicator of how effective a firm's management is.

Asset turnover may be broken into two primary components:

- Working capital management
- Long-term asset management

Working Capital Management

Working capital is the difference between current assets and current liabilities.

Key ratios useful to analysing the management of working capital include:

- Operating working capital to sales
- Operating working capital turnover
- Accounts receivable turnover (day's receivables)
- Inventory turnover (day's inventory)
- Accounts payable turnover (day's payables)

Key Working Capital Ratios

$$\text{Operating working capital to sales ratio} = \frac{\text{Operating working capital}}{\text{Sales}}$$

$$\text{Operating working capital turnover} = \frac{\text{Sales}}{\text{Operating working capital}}$$

$$\text{Accounts receivable turnover} = \frac{\text{Sales}}{\text{Average accounts receivables}}$$

$$\text{Inventory turnover} = \frac{\text{Cost of goods sold}}{\text{Average inventory}}$$

$$\text{Accounts payable turnover} = \frac{\text{Purchases}}{\text{Average accounts payable}} \text{ or } \frac{\text{Cost of goods sold}}{\text{Average accounts payable}}$$

$$\text{Days receivables} = \frac{\text{Average accounts receivable}}{\text{Average sales per day}}$$

$$\text{Days inventory} = \frac{\text{Average inventory}}{\text{Average cost of goods sold per day}}$$

$$\text{Days payables} = \frac{\text{Average accounts payable}}{\text{Average purchases (or cost of goods sold) per day}}$$

Kathmandu's asset management ratios

FIGURE 5.8 Asset management ratios

Ratio	Kathmandu		Super Retail Group		Other peers
	2018	2017	2018	2017	2018
Operating working capital / revenue	9.8%	7.7%	5.1%	6.9%	19.9%
Net non-current operating assets / revenue	71.3%	68.9%	40.6%	39.1%	45.5%
PPE / revenue	12.5%	13.8%	10.4%	10.2%	9.9%
Operating working capital turnover	10.2	12.9	19.7	14.6	5.0

Ratio	Kathmandu		Super Retail Group		Other peers
	2018	2017	2018	2017	2018
Net non-current operating assets turnover	1.4	1.5	2.5	2.6	2.2
PPE turnover	8.0	7.3	9.6	9.8	10.1
Accounts receivables turnover	50.4	78.7	141.7	133.0	68.6
Days' receivables	7.2	4.6	2.6	2.7	5.3
Inventories turnover	1.8	1.8	2.8	2.8	9.3
Days' inventories	201.7	199.2	132.4	131.5	39.3
Accounts payables turnover	2.8	3.1	4.1	4.6	7.3
Days' payable	129.9	116.3	88.9	79.0	49.8
Cash conversion cycle (in days)	79.1	87.5	46.1	55.3	-5.2



Key Takeaways

- Ratio analysis can be used to assess how various line items in FS relate to each other and to measure relative performance.
- Two approaches can be used to decompose profitability: the traditional approach and the alternative approach.
- Ratio analysis must be evaluated while considering whether firm performance is consistent with the strategic initiatives of management.



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See you at the seminar

